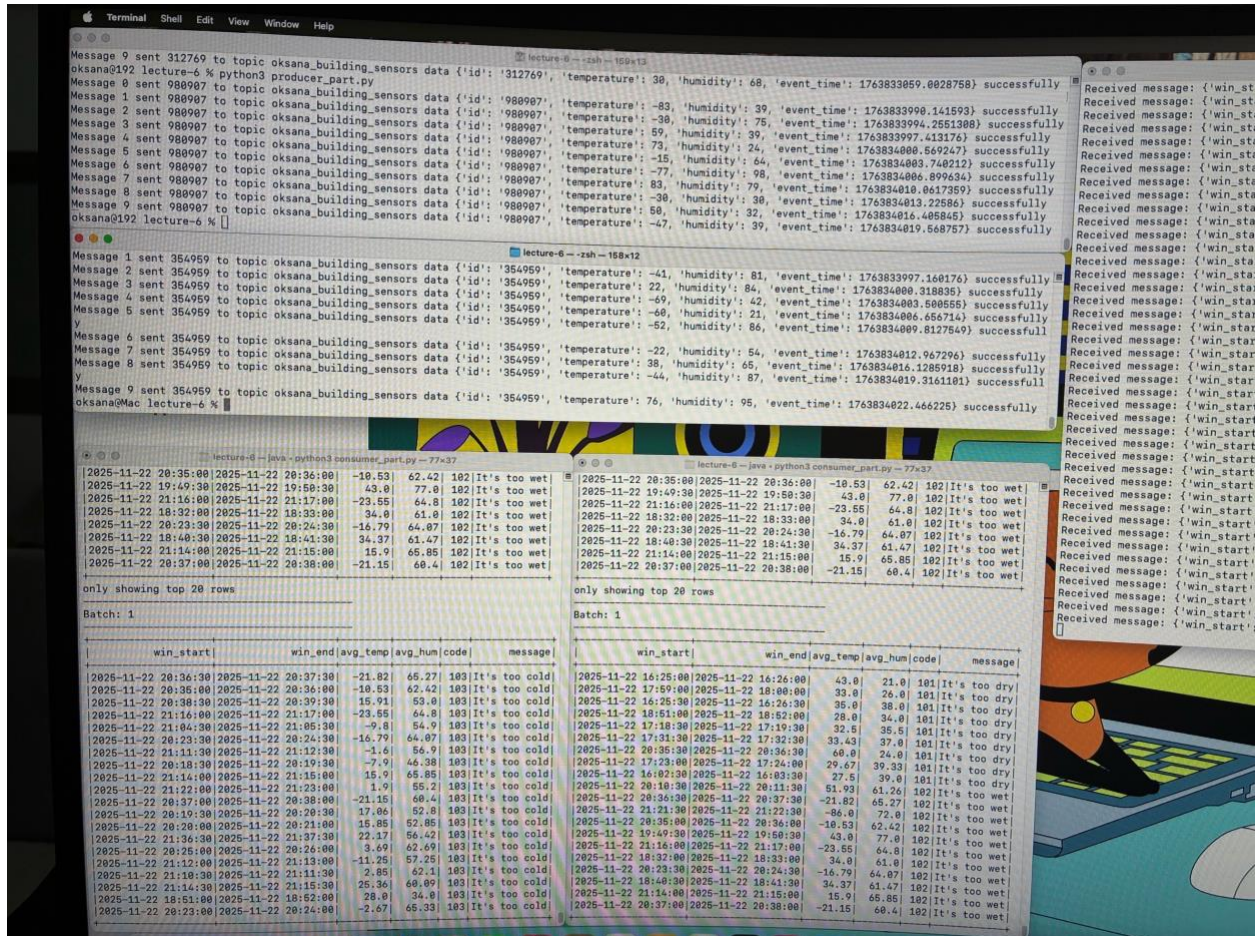


Домашнє завдання до теми «Spark streaming»

1.-4. Генерація потоку даних:



Вхідні данні відредаговано

```
id,humidity_min,humidity_max,temperature_min,temperature_max,code,message
1,0,40,-999,-999,101,"It's too dry"
2,60,100,-999,-999,102,"It's too wet"
3,-999,-999,-30,30,103,"It's too cold"
4,-999,-999,40,100,104,"It's too hot"
```

5. Запис даних у Kafka-топік:

Оскільки завдання неточне, то алерти відсилаються в загальний топік та окремі для вологості і температури. Вичитує стрім Лістелер-скрипт

```
Received message: {'win_start': '2025-11-22T20:24:00.000+04:00', 'win_end': '2025-11-22T20:25:00.000+04:00', 'avg_temp': -13.79, 'avg_hum': 61.76, 'code': 102, 'message': 'It's too w
Received message: {'win_start': '2025-11-22T17:23:00.000+04:00', 'win_end': '2025-11-22T20:24:00.000+04:00', 'avg_temp': -2.67, 'avg_hum': 65.33, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T17:27:00.000+04:00', 'win_end': '2025-11-22T17:28:00.000+04:00', 'avg_temp': 34.0, 'avg_hum': 60.75, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T18:40:30.000+04:00', 'win_end': '2025-11-22T17:41:30.000+04:00', 'avg_temp': 34.37, 'avg_hum': 61.47, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T17:18:30.000+04:00', 'win_end': '2025-11-22T17:19:30.000+04:00', 'avg_temp': 32.5, 'avg_hum': 35.5, 'code': 101, 'message': 'It's too dry'
Received message: {'win_start': '2025-11-22T20:37:00.000+04:00', 'win_end': '2025-11-22T21:15:00.000+04:00', 'avg_temp': 15.9, 'avg_hum': 65.85, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T17:31:30.000+04:00', 'win_end': '2025-11-22T17:32:30.000+04:00', 'avg_temp': -21.15, 'avg_hum': 60.4, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T18:36:00.000+04:00', 'win_end': '2025-11-22T18:37:00.000+04:00', 'avg_temp': 33.43, 'avg_hum': 37.0, 'code': 101, 'message': 'It's too dry'
Received message: {'win_start': '2025-11-22T17:59:00.000+04:00', 'win_end': '2025-11-22T18:00:00.000+04:00', 'avg_temp': 39.33, 'avg_hum': 63.33, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T21:10:00.000+04:00', 'win_end': '2025-11-22T21:11:00.000+04:00', 'avg_temp': 33.0, 'avg_hum': 26.0, 'code': 101, 'message': 'It's too dry'
Received message: {'win_start': '2025-11-22T20:14:00.000+04:00', 'win_end': '2025-11-22T20:15:00.000+04:00', 'avg_temp': -45.0, 'avg_hum': 75.67, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:25:00.000+04:00', 'win_end': '2025-11-22T20:26:00.000+04:00', 'avg_temp': 51.26, 'avg_hum': 67.62, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T21:16:30.000+04:00', 'win_end': '2025-11-22T21:17:30.000+04:00', 'avg_temp': 3.69, 'avg_hum': 62.69, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:10:30.000+04:00', 'win_end': '2025-11-22T20:11:30.000+04:00', 'avg_temp': -22.24, 'avg_hum': 64.82, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:35:30.000+04:00', 'win_end': '2025-11-22T20:36:30.000+04:00', 'avg_temp': 61.93, 'avg_hum': 61.26, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T20:36:30.000+04:00', 'win_end': '2025-11-22T20:37:30.000+04:00', 'avg_temp': 60.0, 'avg_hum': 24.0, 'code': 101, 'message': 'It's too dry'
Received message: {'win_start': '2025-11-22T21:21:30.000+04:00', 'win_end': '2025-11-22T21:22:30.000+04:00', 'avg_temp': -21.82, 'avg_hum': 65.27, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T20:35:00.000+04:00', 'win_end': '2025-11-22T20:36:00.000+04:00', 'avg_temp': -86.0, 'avg_hum': 72.0, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T16:25:00.000+04:00', 'win_end': '2025-11-22T16:26:00.000+04:00', 'avg_temp': -10.53, 'avg_hum': 62.42, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T19:49:30.000+04:00', 'win_end': '2025-11-22T19:50:30.000+04:00', 'avg_temp': 43.0, 'avg_hum': 21.0, 'code': 101, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T21:15:00.000+04:00', 'win_end': '2025-11-22T21:16:00.000+04:00', 'avg_temp': 43.0, 'avg_hum': 77.0, 'code': 102, 'message': 'It's too dry'
Received message: {'win_start': '2025-11-22T18:32:00.000+04:00', 'win_end': '2025-11-22T18:33:00.000+04:00', 'avg_temp': -23.55, 'avg_hum': 64.8, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T21:28:30.000+04:00', 'win_end': '2025-11-22T20:29:30.000+04:00', 'avg_temp': 34.0, 'avg_hum': 61.0, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:23:30.000+04:00', 'win_end': '2025-11-22T20:24:30.000+04:00', 'avg_temp': -11.43, 'avg_hum': 68.29, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:15:00.000+04:00', 'win_end': '2025-11-22T20:16:00.000+04:00', 'avg_temp': -16.79, 'avg_hum': 66.38, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T21:27:30.000+04:00', 'win_end': '2025-11-22T21:28:30.000+04:00', 'avg_temp': 61.25, 'avg_hum': 64.87, 'code': 102, 'message': 'It's too we
Received message: {'win_start': '2025-11-22T21:13:30.000+04:00', 'win_end': '2025-11-22T21:14:30.000+04:00', 'avg_temp': -38.0, 'avg_hum': 62.33, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:14:30.000+04:00', 'win_end': '2025-11-22T20:15:30.000+04:00', 'avg_temp': 4.33, 'avg_hum': 72.89, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T21:37:30.000+04:00', 'win_end': '2025-11-22T21:38:30.000+04:00', 'avg_temp': 60.2, 'avg_hum': 75.6, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T17:23:00.000+04:00', 'win_end': '2025-11-22T17:24:00.000+04:00', 'avg_temp': -15.76, 'avg_hum': 64.75, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:24:30.000+04:00', 'win_end': '2025-11-22T20:25:30.000+04:00', 'avg_temp': 29.67, 'avg_hum': 39.33, 'code': 101, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T21:15:30.000+04:00', 'win_end': '2025-11-22T21:16:30.000+04:00', 'avg_temp': -0.94, 'avg_hum': 61.13, 'code': 102, 'message': 'It's too dry'
Received message: {'win_start': '2025-11-22T21:04:00.000+04:00', 'win_end': '2025-11-22T21:05:00.000+04:00', 'avg_temp': -31.0, 'avg_hum': 64.67, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T21:34:30.000+04:00', 'win_end': '2025-11-22T21:35:30.000+04:00', 'avg_temp': -47.4, 'avg_hum': 60.8, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:33:00.000+04:00', 'win_end': '2025-11-22T19:34:00.000+04:00', 'avg_temp': -10.78, 'avg_hum': 60.78, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T16:02:30.000+04:00', 'win_end': '2025-11-22T16:03:30.000+04:00', 'avg_temp': -12.45, 'avg_hum': 64.97, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T18:37:00.000+04:00', 'win_end': '2025-11-22T18:38:00.000+04:00', 'avg_temp': 27.5, 'avg_hum': 39.0, 'code': 101, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T19:46:30.000+04:00', 'win_end': '2025-11-22T19:47:30.000+04:00', 'avg_temp': 34.5, 'avg_hum': 67.5, 'code': 102, 'message': 'It's too dry'
Received message: {'win_start': '2025-11-22T21:34:00.000+04:00', 'win_end': '2025-11-22T21:35:00.000+04:00', 'avg_temp': 33.75, 'avg_hum': 60.25, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T20:32:30.000+04:00', 'win_end': '2025-11-22T20:33:30.000+04:00', 'avg_temp': -6.4, 'avg_hum': 61.6, 'code': 102, 'message': 'It's too wet'
Received message: {'win_start': '2025-11-22T21:52:30.000+04:00', 'win_end': '2025-11-22T21:53:30.000+04:00', 'avg_temp': 14.36, 'avg_hum': 63.40, 'code': 102, 'message': 'It's too wet'}
```